

Quantum Field Theory I

Homework 1

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Q. 1 - a) For scalar fields, the Lorentz group generators can be written as differential operators acting on functions of space-time coordinates x^μ :

$J^{\mu\nu} = +i(x^\mu \partial^\nu - x^\nu \partial^\mu)$, with the commutator relationship of coordinates and derivations:

$$[\partial^\mu, x^\nu] = g^{\mu\nu}, \quad [x^\nu, \partial^\mu] = -g^{\mu\nu}$$

$$[x^\mu, x^\nu] = 0, \quad [\partial^\nu, \partial^\mu] = 0$$

Let $L^{\mu\nu} \equiv x^\mu \partial^\nu - x^\nu \partial^\mu \Rightarrow J^{\mu\nu} = +i L^{\mu\nu}$, then:

$$\begin{aligned} [L^{\alpha\beta}, L^{\rho\sigma}] &= [x^\alpha \partial^\beta - x^\beta \partial^\alpha, x^\rho \partial^\sigma - x^\sigma \partial^\rho] = \\ &= [x^\alpha \partial^\beta, x^\rho \partial^\sigma] - [x^\alpha \partial^\beta, x^\sigma \partial^\rho] - [x^\beta \partial^\alpha, x^\rho \partial^\sigma] + [x^\beta \partial^\alpha, x^\sigma \partial^\rho] \end{aligned}$$

The Leibniz rule for commutators: $[A, BC] = A[B, C] + B[A, C]$:

$$[A \partial^\mu, B \partial^\nu] = A[\partial^\mu, B] \partial^\nu + B[A, \partial^\nu] \partial^\mu, \text{ then:}$$

$$\begin{aligned} + [x^\alpha \partial^\beta, x^\rho \partial^\sigma] &= x^\alpha [\partial^\beta, x^\rho] \partial^\sigma + x^\rho [x^\alpha, \partial^\sigma] \partial^\beta = \\ &= x^\alpha g^{\beta\rho} \partial^\sigma - x^\rho g^{\alpha\sigma} \partial^\beta = g^{\beta\rho} x^\alpha \partial^\sigma - g^{\alpha\sigma} x^\rho \partial^\beta \\ - [x^\alpha \partial^\beta, x^\sigma \partial^\rho] &= -g^{\beta\sigma} x^\alpha \partial^\rho + g^{\alpha\rho} x^\sigma \partial^\beta \\ - [x^\beta \partial^\alpha, x^\rho \partial^\sigma] &= -g^{\alpha\rho} x^\beta \partial^\sigma + g^{\beta\sigma} x^\rho \partial^\alpha \\ + [x^\beta \partial^\alpha, x^\sigma \partial^\rho] &= g^{\alpha\sigma} x^\beta \partial^\rho - g^{\beta\rho} x^\sigma \partial^\alpha \end{aligned}$$

continues
 $\Rightarrow$

thus:

$$[L^{2P}, L^{P6}] = g^{PP} x^{2P} \partial^6 - g^{26} x^P \partial^P - g^{P6} x^{2P} \partial^P + g^{2P} x^6 \partial^P - g^{2P} x^P \partial^6 + g^{P6} x^P \partial^2 + \dots$$

$$\dots + g^{26} x^P \partial^P - g^{PP} x^6 \partial^2$$

→ grouping by these terms multiplied by the same metric factor:

$$\begin{aligned} \rightarrow g^{PP} (x^{2P} \partial^6 - x^6 \partial^2) &= g^{PP} L^{2P} \\ g^{26} (-x^P \partial^P + x^P \partial^P) &= g^{26} L^{PP} \\ g^{P6} (-x^{2P} \partial^P + x^P \partial^2) &= -g^{P6} L^{2P} \\ g^{2P} (x^6 \partial^P - x^P \partial^6) &= -g^{2P} L^{P6} \end{aligned}$$

$$\Rightarrow [L^{2P}, L^{P6}] = g^{PP} L^{2P} + g^{26} L^{PP} - g^{P6} L^{2P} - g^{2P} L^{P6}$$

$$\rightarrow L^{\mu\nu} = -i J^{\mu\nu} \Rightarrow [J^{2P}, J^{P6}] = -i (g^{PP} J^{2P} + g^{26} J^{PP} - g^{P6} J^{2P} - g^{2P} J^{P6}) \Rightarrow$$

$$\Rightarrow i [J^{2P}, J^{P6}] = g^{PP} J^{2P} + g^{26} J^{PP} - g^{P6} J^{2P} - g^{2P} J^{P6}$$

LL> Because $g^{\mu\mu} = g^{\mu\mu}$, $J^{\mu\mu} = -J^{\mu\mu}$ we can change the indices and find:

$$\underline{i [J^{2P}, J^{P6}] = g^{PP} J^{2P} + g^{26} J^{PP} + g^{PP} J^{26} + g^{26} J^{PP}, Q.E.D.}$$

Q.1-b) The 3 rotation and 3 boost transformations of the Lorentz group can be described as part of $SL(2, \mathbb{C})$ with Lie algebra $\mathfrak{sl}(2, \mathbb{C})$:

- Let J^i be the rotation generators and K^i the boost generators such that:

$$J^i \equiv \frac{1}{2} \epsilon^{ijk} J^{jk}, \quad K^i \equiv J^{0i} \quad \text{where } J^{\mu\nu} \text{ are the six anti-symmetric generators of } \mathfrak{so}(1,3)$$

→ The Lie Algebra of part a) implies then:

$$[J^i, J^j] = i \epsilon^{ijk} J^k, \quad [J^i, K^j] = i \epsilon^{ijk} K^k, \quad [K^i, K^j] = -i \epsilon^{ijk} J^k$$

$$\Rightarrow J^i \equiv \frac{1}{2} \epsilon^{lmn} J^{lm} J^{mn}, \quad J^j \equiv \frac{1}{2} \epsilon^{pq4} J^{pq} \Rightarrow$$

$$[J^i, J^j] = \frac{1}{4} \epsilon^{lmn} \epsilon^{pq4} [J^{lm}, J^{pq}], \quad \text{from part a) } \rightarrow$$

$i [J^{lm}, J^{pq}] = g^{lp} J^{mq} - g^{lp} J^{mq} + g^{mq} J^{lp} - g^{mq} J^{lp}$, because all indices are spatial indices, we can simplify for purely spatial components:

$$i [J^{lm}, J^{pq}] = (-\delta^{lp}) J^{mq} + (-\delta^{lp}) J^{mq} + (-\delta^{mq}) J^{lp} + (-\delta^{mq}) J^{lp}$$

$$\Rightarrow [J^{lm}, J^{pq}] = i (\delta^{lp} J^{mq} - \delta^{lp} J^{mq} - \delta^{mq} J^{lp} + \delta^{mq} J^{lp})$$

↳ knowing that $\epsilon^{lmn} \epsilon^{pq4} = 2 \delta^{lp} \delta^{mq} - 2 \delta^{mq} \delta^{lp}$ we can use it on: $[J^i, J^j] = \frac{1}{4} \epsilon^{lmn} \epsilon^{pq4} [J^{lm}, J^{pq}]$ to find $[J^i, J^j] = i \epsilon^{ijk} J^k$ ✓

→ For $[J^i, K^j]$:

$$[J^i, K^j] = \frac{1}{2} \epsilon^{imn} [J^{mn}, J^{0j}], \text{ from the Lie algebra}$$

$$i[J^{mn}, J^{0j}] = J^{jn} J^{0m} + g^{m0} J^{jn} + J^{mj} J^{0n} + g^{n0} J^{mj} =$$

$$= (-\delta^{jn}) J^{0m} + (-\delta^{mj}) J^{0n} \Rightarrow [J^{mn}, J^{0j}] = i(\delta^{jn} J^{0m} + \delta^{mj} J^{0n})$$

⇒ because the ϵ^{imn} is anti-symmetric in m, n we can contract it to find, with this commutator:

$$\rightarrow [J^i, K^j] = \frac{1}{2} \epsilon^{imn} [J^{mn}, J^{0j}] = i \epsilon^{ijk} K^k \checkmark$$

$$\rightarrow [K^i, K^j] = i[J^{0i}, J^{0j}] = g^{ji} J^{00} + g^{00} J^{ji} + g^{0j} J^{0i} + g^{ji} J^{00}$$

$$g^{ji} = -\delta^{ji}, J^{00} = 0, J^{00} = 1, g^{0i} = g^{0j} = 0$$

$$\Rightarrow [K^i, K^j] = J^{ji} = -J^{ij} = -\epsilon^{ijk} J^k \Rightarrow$$

$$\Rightarrow [K^i, K^j] = -i \epsilon^{ijk} J^k$$

→ To find the finite-dimensional, irreducible representation, we have to complexify this Lie algebra:

$$\text{Let } A^i = \frac{1}{2}(J^i + iK^i), B^i = \frac{1}{2}(J^i - iK^i), \text{ then:}$$

$$[A^i, A^j] = \frac{1}{4} ([J^i, J^j] + i[J^i, K^j] + i[K^i, J^j] - [K^i, K^j])$$

$$= \frac{1}{4} (-i \epsilon^{ijk} J^k + i(\epsilon^{ijk} K^k) + i(\epsilon^{jik} K^k) - (-i \epsilon^{ijk} J^k)) =$$

$$= -\frac{i}{2} \epsilon^{ijk} J^k + \frac{i}{2} \epsilon^{ijk} K^k = +i \epsilon^{ijk} A^k$$

Similarly, we find $[B^i, B^j] = +i \epsilon^{ijk} B^k$, both satisfy

the $\mathfrak{su}(2)_L$ algebra

$$\rightarrow [A^i, B^j] = \frac{1}{4} ([J^i, J^j] - i [J^i, K^j] + i [K^i, J^j] + [K^i, K^j])$$

$$= \frac{1}{4} (-i \epsilon^{ijk} J^k + \epsilon^{ijk} K^k - \epsilon^{ijk} K^k + i \epsilon^{ijk} J^k) = 0$$

$\rightarrow$ In the complex numbers, the Lorentz algebra $\mathfrak{so}(3,1)$ is isomorphic with a direct sum of two commuting $\mathfrak{su}(2)$ algebras:

$$\mathfrak{so}(3,1) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$$

$\hookrightarrow$ And thus can have their irreducible representations given by the tensor product of both $\mathfrak{su}(2)_L$, labeled by a half integer valued j over the dimension

$\hookrightarrow$ the irreducible representation is labeled by the pair (j_1, j_2) of half integers that define the dimension of the tensor product $\dim(j_1, j_2) = (2j_1 + 1)(2j_2 + 1)$

Q.2 - a) Given $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

→ The Euler-Lagrange Equations can be written as:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0, \quad \frac{\partial \mathcal{L}}{\partial A_\nu} = 0$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} &= \frac{\partial}{\partial (\partial_\mu A_\nu)} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) = -\frac{1}{4} \frac{\partial}{\partial (\partial_\mu A_\nu)} (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= -\frac{1}{2} \frac{\partial}{\partial (\partial_\mu A_\nu)} (\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu) = \\ &= -\frac{1}{2} \cdot 2 (\partial^\mu A^\nu - \partial^\nu A^\mu) = -F^{\mu\nu} \end{aligned}$$

⇒ Then the equation of motion is:

$$\partial_\mu F^{\mu\nu} = 0 \quad (\text{or the Maxwell equations in a vacuum})$$

Q.2 - b) $x^\mu \rightarrow \tilde{x}^\mu = x^\mu + a^\mu, \quad a^\mu = \epsilon^\mu \Rightarrow$

$$\Rightarrow A'_\nu(\tilde{x}) = A_\nu(x + \epsilon) = A_\nu(x) + \epsilon^\rho \partial_\rho A_\nu(x) + \mathcal{O}(\epsilon^2) \Rightarrow$$

$$\Rightarrow \delta A_\nu = A'_\nu(\tilde{x}) - A_\nu(x) = \epsilon^\rho \partial_\rho A_\nu$$

→ The Noether current is:

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \delta A_\nu - \mathcal{L} \delta x^\mu = (-F^{\mu\nu}) (\epsilon^\rho \partial_\rho A_\nu) - \mathcal{L} \epsilon^\mu$$

$$\Rightarrow \epsilon^\rho = \eta^{\rho\sigma} \epsilon_\sigma, \quad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \Rightarrow$$

continues

$$\Rightarrow \mathcal{L} = \epsilon_0 (F^{\mu\nu} \partial^\nu A_\mu - \eta^{\mu 0} \mathcal{L}) = \epsilon_0 (F^{\mu\nu} \partial^\nu A_\mu + \frac{1}{4} \eta^{\mu 0} \epsilon_{2\beta} F^{\alpha\beta})$$

$$\Rightarrow \text{making } \epsilon_0 T^{\mu 0} = \mathcal{L} :$$

$$T^{\mu 0} = F^{\mu\nu} \partial^\nu A_\mu + \frac{1}{4} \eta^{\mu 0} F_{2\beta} F^{\alpha\beta}$$

Q.3) Let's consider the transformation

$$x'^\mu = e^\epsilon x^\mu, \quad \phi'(x) = e^{-\epsilon} \phi(x), \quad d\phi = 1, \quad \text{then } \rightarrow$$

$$x^\mu = e^{-\epsilon} x'^\mu \rightarrow \phi'(x) = e^{-\epsilon} \phi(e^\epsilon x). \quad \text{To compute}$$

$$\rightarrow \partial_\mu \phi \partial^\mu \phi :$$

$$\partial_\mu \phi' = \frac{\partial}{\partial x^\mu} [e^{-\epsilon} \phi(e^\epsilon x)] = e^{-2\epsilon} (\partial_\mu \phi)(e^\epsilon x)$$

$$\partial^\mu \phi' = \eta^{\mu\nu} \partial_\nu \phi'(x) = e^{-2\epsilon} (\partial^\mu \phi)(e^\epsilon x)$$

$$\Rightarrow \partial_\mu \phi' \partial^\mu \phi' = e^{-4\epsilon} [\partial_\mu \phi \partial^\mu \phi](e^\epsilon x)$$

$$\rightarrow \text{For the potential term: } [\phi'(x)]^4 = e^{-4\epsilon} \phi^4(e^\epsilon x)$$

Thus the transformed Lagrangian is:

$$\mathcal{L}' = \frac{1}{2} \partial_\mu \phi' \partial^\mu \phi' - \frac{\lambda}{4!} [\phi'(x)]^4 = e^{-4\epsilon} \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right]_{x \rightarrow e^\epsilon x}$$

$$= e^{-4\epsilon} \mathcal{L}(e^\epsilon x)$$

→ For it to be a symmetry, the action must be the same.

$$S' = \int d^4x \mathcal{L}'(x) = \int d^4x \tilde{\epsilon}^{-4\epsilon} \mathcal{L}(\tilde{\epsilon}^{-\epsilon} x), \quad \text{making } \gamma^\mu = \tilde{\epsilon}^{-\epsilon} x^\mu$$

$$d^4x = \tilde{\epsilon}^{+4\epsilon} d^4y$$

$$\Rightarrow S' = \int d^4x \mathcal{L}'(x) = \int d^4y \tilde{\epsilon}^{-4\epsilon} \tilde{\epsilon}^{+4\epsilon} \mathcal{L}(y) = \int d^4y \mathcal{L}(y) = S$$

↳ Then the transformation is a symmetry

⇒ The conserved current is given by:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi + \mathcal{L} \delta x^\mu$$

↳ We can expand in first order so that:

$$\tilde{\epsilon}^{-\epsilon} = 1 - \epsilon + \mathcal{O}(\epsilon^2) \quad f(x - \epsilon x) = f(x) - \epsilon x^\nu \partial_\nu f(x) + \mathcal{O}(\epsilon^2)$$

$$\Rightarrow \phi'(x) = (1 - \epsilon) [\phi(x) - \epsilon x^\nu \partial_\nu \phi(x)] + \mathcal{O}(\epsilon^2)$$

$$= \phi(x) - \epsilon \phi(x) - \epsilon x^\nu \partial_\nu \phi(x) + \mathcal{O}(\epsilon^2) \Rightarrow$$

$$\Rightarrow \delta \phi(x) \equiv \phi'(x) - \phi(x) = -\epsilon (\phi + x^\nu \partial_\nu \phi)$$

$$\delta x^\mu \equiv x'^\mu - x^\mu = \epsilon x^\mu$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = j^\mu \phi$$

continue
⇒

$$\partial_r F^{\mu\nu}$$

$$\sum \partial_i F_i$$



$$\begin{aligned} \gamma^\mu &= \partial^\mu \phi [-\epsilon (\phi + x^\nu \partial_\nu \phi)] + 2(\epsilon x^\mu) = \\ &= -\epsilon [\phi \partial^\mu \phi + x^\nu \partial^\mu \phi \partial_\nu \phi - x^\mu \partial^\mu \phi] \end{aligned}$$

Let's call $T_\mu^\nu = \partial^\nu \phi \partial_\mu \phi - \delta_\mu^\nu \mathcal{L}$, and let's

$D^\mu = x^\nu T_\nu^\mu + \phi \partial^\mu \phi \Rightarrow \gamma^\mu = -\epsilon D^\mu$ or yet, because ϵ is an arbitrary parameter, the conservation of the current becomes:

$$\partial_\mu D^\mu = 0, \quad D^\mu = x^\nu T_\nu^\mu + \phi \partial^\mu \phi, \quad T_\mu^\nu = \partial^\nu \phi \partial_\mu \phi - \delta_\mu^\nu \mathcal{L}$$

Q.4a) Calling the metric $\eta = (\eta_{11}, \eta_{22})$, the Lorentz transformation is invariant:

$$\int_{\mu\nu} \Lambda_\mu^\alpha \Lambda_\nu^\beta = g_{\alpha\beta}$$

Then for an infinitesimal transformation:

$\Lambda_\mu^\alpha = \delta_\mu^\alpha + \delta\omega_\mu^\alpha$, where $|\delta\omega| \ll 1$, then the invariance at first order:

$$\int_{\mu\nu} (\delta_\mu^\alpha + \delta\omega_\mu^\alpha) (\delta_\nu^\beta + \delta\omega_\nu^\beta) = g_{\alpha\beta} + g_{\mu\nu} \delta\omega_\mu^\alpha \delta\omega_\nu^\beta + g_{\mu\nu} \delta\omega_\mu^\alpha \delta_\nu^\beta + g_{\mu\nu} \delta_\mu^\alpha \delta\omega_\nu^\beta + \mathcal{O}(\delta\omega^2)$$

$$\Rightarrow g_{\alpha\beta} + g_{\mu\nu} \delta\omega_\mu^\alpha \delta\omega_\nu^\beta + g_{\mu\nu} \delta\omega_\mu^\alpha \delta_\nu^\beta + g_{\mu\nu} \delta_\mu^\alpha \delta\omega_\nu^\beta = g_{\alpha\beta}$$

$$\Rightarrow g_{\alpha\beta} \delta\omega_\mu^\alpha \delta\omega_\nu^\beta + g_{\mu\nu} \delta\omega_\mu^\alpha \delta_\nu^\beta = 0$$

continued

=>

tilibra

$$\begin{aligned} &\rightarrow \text{Via the metric } \delta w_{\alpha\beta} = g_{\alpha\gamma} \delta w^\gamma_\beta \\ &\Rightarrow g_{\alpha\gamma} \delta w^\gamma_\beta + g_{\mu\beta} \delta w^\mu_\alpha = 0 \Rightarrow \delta w_{\alpha\beta} + \delta w_{\beta\alpha} = 0 \\ &\Rightarrow \underline{\delta w_{\alpha\beta} = -\delta w_{\beta\alpha}} \quad // \quad \text{Q.E.D.} \end{aligned}$$

b) Rotations are a mix of space coordinates,
for 3D Euclidean space:

$$\delta x^0 = 0, \quad \delta x^i = \epsilon^i_{jk} \theta^j x^k, \quad \text{for a general rotation transformation:}$$

$$\delta x^\mu = \delta w^\mu_\nu x^\nu \rightarrow \delta w^0_0 = \delta w^0_i = \delta w^i_0 = 0$$

$$\Rightarrow \delta w^i_k = \epsilon^i_{jk} \theta^j, \quad \text{for spatial coordinates:}$$

$$g_{ij} = -\delta_{ij}, \quad \delta w_{\mu\nu} = g_{\mu\rho} \delta w^\rho_\nu \Rightarrow$$

$$\delta w_{ij} = g_{i\rho} \delta w^\rho_j = (-\delta_{ik}) \epsilon^k_{\ell j} \theta^\ell = -\epsilon^i_{\ell j} \delta w^\ell$$

$$= \epsilon_{i\ell j} \delta\theta^\ell = \epsilon_{ijk} \delta\theta^k$$

Q.5) given $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

$$x'^\mu = \Lambda^\mu_\nu x^\nu, A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x)$$

↳ Let's do infinitesimal transformation:

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \delta\omega^\mu_\nu, \text{ considering } \delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu} \Rightarrow$$

$\Rightarrow \delta x^\mu \equiv x'^\mu - x^\mu = \delta\omega^\mu_\nu x^\nu$, then for the transformed field:

$$A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x) + \delta\omega^\mu_\nu A^\nu(x)$$

$\Rightarrow$ Expanding in first order $\delta\omega$:

$$A'^\mu(x') = A'^\mu(x + \delta x) \approx A'^\mu(x) + \delta x^\nu \partial_\nu A'^\mu(x) \approx A'^\mu(x) + \delta\omega^\mu_\nu x^\nu \partial_\nu A^\mu(x) \Rightarrow$$

$$\Rightarrow A'^\mu(x) + \delta\omega^\mu_\nu x^\nu \partial_\nu A^\mu = \Lambda^\mu_\nu A^\nu + \delta\omega^\mu_\nu A^\nu(x)$$

Then the variation of the field:

$$\Delta A^\mu(x) \equiv A'^\mu(x) - A^\mu(x) = \delta\omega^\mu_\nu A^\nu(x) - \delta\omega^\mu_\nu x^\nu \partial_\nu A^\mu(x)$$

$\Rightarrow$ For the Lagrangian density:

$$\mathcal{L}'(x) = \mathcal{L}(x) \Rightarrow \mathcal{L}'(x) = \mathcal{L}(x - \delta x) \approx \mathcal{L}(x) - \delta x^\mu \partial_\mu \mathcal{L}(x)$$

$$\Rightarrow \Delta \mathcal{L} \equiv \mathcal{L}'(x) - \mathcal{L}(x) = -\delta\omega^\mu_\nu x^\nu \partial_\mu \mathcal{L}$$

$$\hookrightarrow \text{for } \delta\omega^\mu_\nu = 0 \Rightarrow \Delta \mathcal{L} = \partial_\mu (-\delta\omega^\mu_\nu x^\nu \mathcal{L}) \equiv \partial_\mu \Omega^\mu$$

continues

$\Rightarrow$

$$\Rightarrow \Omega^\mu = -\delta\omega_r^\mu x^2$$

↳ Now, then, the Noether current.

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_2)} \Delta A^\mu = \Omega^\mu, \text{ then}$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_2)} = -F^{\mu 2} \Rightarrow j^\mu = -F^{\mu 2} (\delta\omega_r^\mu A_2 - \delta\omega_r^\mu x^2 A_2) + \delta\omega_r^\mu x^2$$

↳ it's interesting using $\delta\omega_{\alpha\beta} = \frac{1}{2} \delta\omega_{\rho\sigma} (\delta_\alpha^\rho \delta_\beta^\sigma - \delta_\alpha^\sigma \delta_\beta^\rho)$

$$\Rightarrow \delta\omega_r^\mu = \frac{1}{2} \delta\omega_{\rho\sigma} (y^{\rho\mu} \delta_r^\sigma - y^{\sigma\mu} \delta_r^\rho)$$

$$\delta\omega_r^\mu = \frac{1}{2} \delta\omega_{\rho\sigma} (y^{\mu\rho} \delta_r^\sigma - y^{\mu\sigma} \delta_r^\rho), \text{ we can write}$$

$$j^\mu = \frac{1}{2} \delta\omega_{\rho\sigma} M^{\mu, \rho\sigma}, \text{ where } M^{\mu, \rho\sigma} \text{ is defined:}$$

$$M^{\mu, \rho\sigma} = - (F^{\mu\rho} A^\sigma - F^{\mu\sigma} A^\rho) + F^{\mu 2} (x^\rho \partial^\sigma A_2 - x^\sigma \partial^\rho A_2) \\ = (y^{\mu\rho} x^\sigma - y^{\mu\sigma} x^\rho) 2$$